

SPN 521200/FMI 4 – SSAM02T

Utility Light 1 – Voltage Below Normal

Table 1, SPN 521200/FMI 4

Description	Utility light 1 voltage is below normal or is shorted to a low source.
Monitored Parameter	N/A
Enable Conditions	The SSAM detects voltage below normal on connector 9, pin 20.
Monitored Sequence	Continuous
Execution Frequency	Always
Typical Duration	2 seconds
HMI Indication	None
Symptom	Utility light lamps are inoperative.
Verification	Utility light lamps are operational.

Table 1, SPN 521200/FMI 4

Table 2, Wiring

Description	ECU	Connector/Pin	Component	Connector/Pin	Module	Circuit
UTILITY LIGHT #1	SSAM02T	9/20	UTILITY LAMP 1	PWR/A	–	–
UTILITY LIGHT #1	SSAM02T	9/20	UTILITY LAMP 1	GND/B	–	–
UTILITY LIGHT #1	SSAM02T	9/20	UTILITY LAMP 2	GND/B	–	–
UTILITY LIGHT #1	SSAM02T	9/20	UTILITY LAMP 2	PWR/A	–	–

Table 2, Wiring

Fault Code Troubleshooting

1. ☐ With the battery switch in the ON position, turn the key ON and connect to DiagnosticLink®. Are faults for low voltage being reported by other modules?

- a.

☐ **YES** → Repair as needed. Verify the repair.
- b.

☐ **NO** → Go to step 2.
2.

☐ Verify that the vehicle is correctly parameterized. Is the vehicle correctly parameterized?

a.

☐ **YES** → Go to step 3.

b.

☐ **NO** → Correct the vehicle parameters, then verify the repair.

3.

☐ Check voltage between cavity A at the tractor utility lamp plug, and battery positive (+). Is voltage present?

a.

☐ **YES** → Go to step 4.

b.

☐ **NO** → Clear the fault, then verify the repair.

4.

☐ Disconnect Single Signal and Actuation Module (SSAM) connector 9. Check the voltage between pin 20 (harness side), and battery positive (+). Is voltage present?

a.

☐ **YES** → Repair the short to ground in the wiring harness, then verify the repair.

b.

☐ **NO** → Go to step 5.

5.

☐ Perform a fixed load voltage drop test on the circuit. See [Fixed-Load Voltage Drop Test](#). Does the circuit pass the test?

a.

☐ **YES** → Install a test SSAM and attempt to clear the fault. If the fault clears, the vehicle SSAM may be faulty.

▪ [Replacement of the Single SAM](#) in the *New Cascadia Workshop Manual*

▪ [Replacement of the Single SAM](#) in the *47X & 49X Workshop Manual*

▪ [Replacement of the ASAM](#) in the *57X Workshop Manual*

b.

☐ **NO** → Repair the circuit, then verify the repair.

NOTE:

Component Code / VMRS = 034-004-184 — LAMP, UTILITY

Component Code / VMRS = 034-004-287 — WIRING HARNESS, UTILITY LAMP

Component Code / VMRS = 037-001-012 — MODULE, SIGNAL ACTIVATION (SAM), SINGLE

Note:
F07,F15,F16,F18,F25,F26,F35,F36,F41,F42,F43,F44,W12,W13,W14,W15,W16,W17,W18
SPN521200/FMI4
SSAM02T
TROUBLESHOOT
034-004-287,034-004-184,037-001-012

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